Lecture - 6

Statistical Science

Design	Description	Inference
How we collect the data?	summarize the data	using sample to understand popula^n
survey, expt., sampling, etc.	$\bar{x}$, s^2, s, median, etc	

- Population is the full collec^n of items that we are interested in. A sample is a subset of popula^n for which we actually observe data. We calc. sample's mean ht $\bar{X}$ to estimate μ (true popula^n mean).
- Population $\rightarrow \mu, \sigma^2, \sigma, p$ (proportion)
  sample $\rightarrow \bar{X}, s^2, s, \hat{p}$
  A parameter describes the popula^n and is fixed. Statistic is calculated from random sample and can vary. Once the sample is random, statistic becomes a RV.
- Sampling distribution is the probability distribution of a statistic over all possible samples of a particular size

$$X_1 \ldots X_{50} \Rightarrow \underbrace{\bar{X}_1, \bar{X}_2, \ldots \bar{X}_k}_{\text{these are sample means}}$$

sampling distribu^n of $\bar{X}$. $\leftarrow$ distribution

- popula^n distribu^n $\Rightarrow$ distribu^n of X for indi.
- Let $X_1, \ldots X_n$ be indep. RV, each with probab. distribu^n $f(x)$. Then $X_1, \ldots X_n$ form a random sample of size n.

$$f(x_1, x_2 \ldots x_n) = f(x_1)\, f(x_2) \ldots f(x_n)$$

$X_1 \ldots X_n$ are iid $\rightarrow$ independent & identically distributed

- $X_1, \dots, X_n$ is a random sample from a popula^n with $E[X_i] = \mu$ and $Var(X_i) = \sigma^2$.

Sample mean, $\boxed{\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i}$

depends on random obs., it is random.

We want, $E[\bar{X}]$ & $Var(\bar{X})$

- mean of sampling distribu^n

$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} \Rightarrow E[\bar{X}] = E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right]$

$E[\bar{X}] = \frac{1}{n}\sum_{i=1}^{n} E[X_i]$ by linearity of expecta^n

every X_i has mean $\mu \Rightarrow E[\bar{X}] = \frac{1}{\cancel{n}}(\cancel{n}\mu)$

$\boxed{E[\bar{X}] = \mu}$ → sample mean is centered at the popula^n mean. $\bar{X}$ is an unbiased estimator of μ.

- var. of sampling distribu^n

$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$, $Var(\bar{X}) = Var\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right)$

$Var(\bar{X}) = \frac{1}{n^2} Var\left(\sum_{i=1}^{n} X_i\right)$

if the obs. are indep. $Var(\Sigma X_i) = \sum_i Var(X_i)$

$Var(\bar{X}) = \frac{1}{n^{\cancel{2}}}(\cancel{n}\sigma^2) \Rightarrow \boxed{Var(\bar{X}) = \frac{\sigma^2}{n}}$

$\boxed{SD(\bar{X}) = SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}}$

→ stand. error of the mean

- as $n \uparrow$, $\sqrt{n} \uparrow$, so, $\sigma/\sqrt{n} \downarrow$. This means that large samples gives sample mean that tends to lie closer to the popula^n mean.

Eg. 6 rats. $\mu = \frac{w_1 + \dots + w_6}{6} = \frac{84}{6} = 14$

if $n = 2$, ${}^6C_2 = 15$ samples

$E[\bar{X}] = 14$

we get the same result for $n = 5$

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- what we can observe, $n=2$, sample mean is spread more widely, $n=5$, they cluster close to 14.

$\boxed{\text{larger } n \Rightarrow \text{smaller sampling error}}$

- Sampling error arises from the diff. because we use sample and not popula^n. given by $\bar{X}-\mu$.
So sample variability is unavoidable. But with large sample, it reduces.
- $X_i \sim N(\mu, \sigma^2)$, $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$
size n

$$E[\bar{X}] = \mu, \quad Var(\bar{X}) = \frac{\sigma^2}{n}, \quad SD(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

- For any iid popula^n with finite mean & var. $E[\bar{X}] = \mu$ & $Var(\bar{X}) = \sigma^2/n$. If the popula^n is normal, $X_i \sim N(\mu, \sigma^2)$, then $\bar{X}$ is normal, even for small n.
- CLT (Central Limit Th.)
if $X_1 \ldots X_n$ are iid with mean μ & var $\sigma^2 < \infty$, then for sufficiently large n, $\bar{X}$ is approx. normal.

$$\boxed{\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)}$$ even if the original popula^n is not normal.

- Formally, $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$ as $n \to \infty$.

$$\boxed{Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}} \quad \text{and } Z \approx N(0,1)$$

- So CLT says that normality can emerge from averaging. If popula^n is skewed & n is small, $\bar{X}$ may not be normal. As $n \uparrow$, distribution of $\bar{X}$ becomes more normal

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$X \sim N(\mu, \sigma^2) \Rightarrow \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

for every n.

for large 'n', $\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$ } CLT.

Eg. $\mu = 125$, $\sigma = 15$, $n = 40$

$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$,

$SE = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{40}} = \frac{15}{6.32}$

$= 2.37$

(a) $P(120 < \bar{X} < 130)$

$z_1 = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{120 - 125}{2.37} = -2.108$

$z_2 = \frac{130 - 125}{2.37} = 2.108$

$P(120 < \bar{X} < 130) = P(-2.108 < z < 2.108)$

$\phi(2.108) = 0.9825$

$\phi(-2.108) = 1 - \phi(2.108) = 0.0175$

$P(-2.108 < z < 2.108) = 0.9825 - 0.0175$

≈ 0.965

so, there is 96.50% probab. that mean wt. of 40 randomly selected baby giraffe is b/w. 120 & 130 pounds.

(b) find 75th percentile of sample.

$P(\bar{X} < x) = 0.75$ $\qquad z_{0.75} = 0.6745$

$z = \frac{x - \mu}{\sigma/\sqrt{n}} \Rightarrow 0.6745 = \frac{x - 125}{2.37}$

$x \approx 126.60$

so 75th percentile is 126.60 pounds.

- Let's say each obs. is a success/failure outcome. Let p be the true population proportion if we have X success in a sample of size n,

then $\boxed{\hat{p} = \frac{X}{n}}$

$E[\hat{p}] = p$. $\hat{p}$ is an unbiased estimator of the populan. proportion.

$Var(\hat{p}) = \frac{p(1-p)}{n}$ $SD(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$

Because $\hat{p}$ is just a sample mean of bernoulli variables.

$Z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$ analogous to sample mean formula

$np \geq 15$ & $np(1-p) \geq 15$

$\hat{p} \approx N\left(p, \frac{p(1-p)}{n}\right)$